

AYUSH PALLOD

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M.Tech ML researcher & AI engineer — VLM faithfulness, grounding regularisation, and production-grade LLM systems.

EDUCATION

Timeline	Degree & Specialisation	Institute	CGPA
2025–2027	M.Tech in IT (ML & Intelligent Systems)	Indian Institute of Information Technology Allahabad	9.43
2021–2025	B.E. in Computer Science & Engineering	Osmania University, Matrusri Engineering College	8.72

SKILLS AND TOOLS

- **Programming:** Python, C, C++, SQL
- **ML / DL & Agentic:** PyTorch, TensorFlow, Hugging Face (Transformers), PEFT/LoRA, Grounding DINO, SAM, LangChain, LangGraph
- **Eval & Observability:** RAGAS (faithfulness, context precision/recall), CHAIR, POPE, TAM-IoU, Weights & Biases, MLflow, Evidently AI
- **MLOps & Infra:** Docker, GitHub Actions CI/CD, FastAPI, Redis, Celery, PostgreSQL, Qdrant, Ollama, NVIDIA L4/T4 GPUs, E2E Networks, Linux

RESEARCH

VLM Faithfulness via Attention-Grounding Regularisation *Current Research, CVBL, IIIT Allahabad* **2025–2027**

- Reframed **object hallucination** as a **faithfulness failure** — outputs diverge from visual evidence despite correct input, a tractable proxy for misaligned generation; fine-tuned **Qwen2-VL-2B with LoRA** using a joint cross-entropy + **grounding regularisation loss** (Grounding DINO + SAM).
- **~47% CHAIR reduction, ~10% POPE precision improvement**; identified that metric gains may reflect output suppression rather than genuine grounding — informing cleaner alignment-relevant diagnostics.

Causality-Guided Zero-Shot Anomaly Detection *M.Tech Thesis, IIIT Allahabad* **2025–2027**

- **Causal deep learning framework** for multivariate time-series anomaly detection; **<26ms inference latency** and robust generalisation across $100\times$ domain shifts.
- Directly applicable to monitoring deployed AI systems for distributional drift — bridging anomaly detection and alignment-relevant runtime oversight.

PROJECTS

FinSight AI — Enterprise Financial Intelligence Platform [[GitHub](#)] *FastAPI, Qdrant, PostgreSQL, Redis, Celery, Ollama, Docker, Prometheus, Grafana* **2026**

- **Production-grade hybrid RAG platform** (BM25 + BGE-M3 + RRF) with rigorous **RAGAS evaluation** (faithfulness, context precision/recall, answer relevancy) and **prompt-injection detection + PII redaction** guardrails.
- **9-microservice Docker Compose architecture** with async ingestion/OCR/embedding pipelines; **Prometheus/Grafana** p50/p95/p99 observability; automated **GitHub Actions CI/CD**.

AI Multi-Agent Stock Analyst Bot [[GitHub](#)] *LangGraph, RAG, Ollama, Streamlit* **2026**

- **LangGraph DAG-based multi-agent system** with specialized agents (technical, sentiment, fundamental) + manager synthesizer; **automated out-of-scope query rejection** to constrain agent behaviour within intended scope.
- Self fine-tuned **LLaMA 3.2** domain-specific LLM deployed via Ollama; **RAG-grounded execution** to suppress confabulation across agent responses.

Stochastic Finance & Valuation Platform [[GitHub](#)] [[Live](#)] *Numba JIT, ARIMA, Monte Carlo, GitHub Actions* **2026**

- **50 $\times$ execution speedup** via Numba JIT compilation on nested Monte Carlo pathways; multi-asset GBM and ARIMA ensemble forecasting with DCF valuation.
- Deployed live on Streamlit Cloud with **GitHub Actions CI** validating GBM, JIT numerical equivalence, and DCF logic end-to-end.

EXPERIENCE

Amptronics Systems Pvt. Ltd. | Data Analyst Intern **May–June 2024**

- Engineered **real-time data pipelines** for WBTMS telemetry; optimised relational queries across Azure and SQLite architectures.
- Automated end-to-end operational workflows based on client requirements, eliminating manual overhead by **~35%**.

Slash Mark | Data Analyst Intern **Nov 2023–Feb 2024**

- Analysed large-scale multivariate datasets using Pandas, Matplotlib, and Seaborn; produced visualisation dashboards isolating critical trends to deliver **actionable asset insights**.

CERTIFICATIONS & ACHIEVEMENTS

- **Winner**, OOC Hackathon 2026 — BTC quantitative backtesting engine, IIIT Allahabad. **2026**
- **Top 2% Nationwide**, GATE (CS/IT) — normalised score 593. **2025**
- **Innovative Project Award**, CSI Hyderabad Chapter — Brain Tumour Detection Framework. **2024**
- **Machine Learning Specialization** — Andrew Ng, Coursera (Stanford / DeepLearning.AI).
- **MLOps Zoomcamp Certification** — DataTalks.Club.